

Topic C11

Organic chemistry

Concept guide · Worked examples · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, chemical reasoning, worked applications, common misconceptions and original exam-style practice for this topic.

Syllabus map

c11.1	Formulae, names and structures
c11.2	Fuels and petroleum
c11.3	Homologous series
c11.4	Alkanes
c11.5	Alkenes
c11.6	Alcohols
c11.7	Carboxylic acids
c11.8	Polymers

How to study: retrieve definitions and equations, cover the worked examples, then complete the practice paper without using the notes.

1. Representing organic compounds

A molecular formula gives numbers of each atom; a displayed formula shows every atom and bond; a structural formula shows connectivity in a compact form. Members of a homologous series share a functional group and general formula, have similar chemical properties, show gradual physical-property trends and differ successively by CH_2 . Isomers share a molecular formula but have different structures.

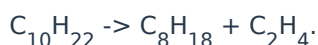
Examiner insight

Name the longest carbon chain and identify the functional group or multiple bond.

2. Petroleum, fuels and cracking

Petroleum is a mixture of hydrocarbons separated by fractional distillation. Fractions condense over temperature ranges; smaller molecules have lower boiling points, are more volatile and ignite more easily. Cracking uses heat and a catalyst to convert long-chain alkanes into shorter alkanes and alkenes, matching demand and supplying monomers.

Worked example



Examiner insight

Fractional distillation is physical separation; cracking is a chemical reaction.

3. Alkanes and combustion

Alkanes are saturated hydrocarbons with general formula $\text{C}_n\text{H}_{2n+2}$. Complete combustion forms carbon dioxide and water; incomplete combustion forms carbon monoxide and/or carbon. Alkanes undergo substitution with chlorine in ultraviolet light, replacing hydrogen by chlorine.

Worked example



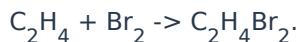
Examiner insight

“Saturated” means carbon-carbon single bonds only, not that the substance contains water.

4. Alkenes

Alkenes are unsaturated hydrocarbons containing C=C and have general formula C_nH_{2n} . They undergo addition reactions. Bromine water changes orange-brown to colourless, distinguishing an alkene from an alkane. Hydrogen forms an alkane; steam forms an alcohol.

Worked example



Examiner insight

Addition uses the double bond and produces one saturated product.

5. Alcohols and carboxylic acids

Ethanol can be made by fermentation of glucose using yeast at about 30-40 °C without oxygen, or by hydration of ethene using steam and a catalyst. Alcohols combust and can oxidise to carboxylic acids. Ethanoic acid reacts as an acid and esterifies with ethanol to form ethyl ethanoate and water.

Worked example

glucose $\rightarrow$ 2ethanol + 2carbon dioxide; ethanol + ethanoic acid $\rightarrow$ ethyl ethanoate + water.

Examiner insight

Compare fermentation, which is renewable, dilute and slow, with hydration, which is fast, continuous and uses a fossil feedstock.

6. Polymers

Addition polymerisation joins many alkene monomers and forms only the polymer; the C=C becomes C-C in the repeat unit. Condensation polymerisation joins bifunctional monomers and eliminates a small molecule such as water. Polyesters contain ester links; proteins are natural condensation polymers of amino acids.

Examiner insight

Keep continuation bonds through the brackets of a polymer repeat unit and include the subscript n.

Exam technique

High-value habits

Use precise chemical vocabulary, balance equations, show every calculation step, include units and state symbols where required, distinguish observations from conclusions, and answer the command word directly.

Retrieval checklist

- Recall all definitions, tests and equations without notes.
- Explain one unfamiliar observation using particles, bonding or collision theory.
- Check charge, atoms, units and significant figures in every calculation.
- Complete one timed set and mark it only after finishing.

Practice Paper

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Instructions

Answer all questions. Select one option for multiple choice. Show working in open spaces and write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A · Multiple choice

Choose the best answer for each question.

1. Successive members of a homologous series differ by [1]
 - A CH
 - B CH₂
 - C C₂H₂
 - D OH

2. Which is an alkane? [1]
 - A C₃H₆
 - B C₄H₈
 - C C₄H₁₀
 - D C₂H₄

3. Which property increases with hydrocarbon chain length? [1]
 - A volatility
 - B ease of ignition
 - C boiling point
 - D flammability

4. Cracking produces [1]
 - A only long alkanes
 - B shorter alkanes and alkenes
 - C metals and oxygen
 - D alcohols only

5. Complete combustion of ethanol forms [1]
 - A CO and H₂
 - B CO₂ and H₂O
 - C C and H₂
 - D CH₄ and O₂

6. Bromine water is decolourised by [1]
 - A ethane
 - B propane
 - C ethene
 - D methane

7. Hydration of ethene produces [1]
 - A ethanol
 - B ethanoic acid
 - C ethyl ethanoate
 - D poly(ethene)

8. Fermentation requires [1]
- A oxygen and 100 °C
 - B yeast and absence of oxygen
 - C a metal catalyst at high pressure only
 - D chlorine and ultraviolet
9. Ethanol oxidises to [1]
- A ethene
 - B ethanoic acid
 - C methane
 - D polyester
10. Addition polymerisation of ethene produces [1]
- A poly(ethene)
 - B polyester
 - C protein
 - D ethanol

Section B · Structured questions

11 The hydrocarbons ethane, ethene and propane are compared.

(a) Give the molecular formula of each. [3]

(b) Identify the unsaturated compound. [1]

(c) Describe a chemical test and its results. [2]

(d) Define structural isomers. [2]

[Total: 8]

12 Petroleum is separated into fractions.

(a) Explain the basis of fractional distillation. [2]

(b) State two trends as chain length increases. [2]

(c) Explain why cracking is useful. [2]

(d) Write one balanced cracking equation. [2]

[Total: 8]

13 Ethanol is produced by fermentation and by hydration of ethene.

(a) State conditions for fermentation. [2]

(b) Write the fermentation equation. [2]

(c) State two advantages of hydration. [2]

(d) State two advantages of fermentation. [2]

[Total: 8]

14 Ethene reacts with bromine and can form poly(ethene).

(a) Describe the bromine reaction. [2]

(b) Name the reaction type. [1]

(c) Explain how the C=C changes during polymerisation. [2]

(d) Describe the repeat unit, including continuation bonds and n. [3]

[Total: 8]

15 Ethanoic acid reacts with ethanol.

(a) Name the organic product. [1]

(b) Name the other product. [1]

(c) Name the reaction type and catalyst. [2]

(d) State two characteristic acid reactions of ethanoic acid. [2]

(e) Explain why the reaction is reversible. [2]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award method and chemistry marks where shown, then identify the first point at which an answer becomes incorrect.

Solutions · Multiple choice

- 1 B. Each successive chain has one extra carbon and two hydrogens.
- 2 C. Alkanes follow C_nH_{2n+2} .
- 3 C. Stronger intermolecular forces raise boiling point.
- 4 B. Large hydrocarbons split into smaller useful molecules.
- 5 B. Complete oxidation yields carbon dioxide and water.
- 6 C. The alkene adds bromine across $C=C$.
- 7 A. Steam adds across the double bond.
- 8 B. Yeast enzymes convert glucose anaerobically.
- 9 B. The alcohol functional group is oxidised to a carboxylic acid.
- 10 A. Ethene monomers join through opened double bonds.

Solutions · Structured questions

11. 11(a) C_2H_6 , C_2H_4 and C_3H_8 [3]. (b) Ethene [1]. (c) Bromine water: ethene changes orange-brown to colourless; alkanes show no change [2]. (d) Same molecular formula but different structural formulae [2].

12. 12(a) Different boiling ranges, with vaporisation and condensation in a column [2]. (b) Boiling point and viscosity increase; volatility and flammability decrease [any 2]. (c) Converts surplus long chains to demanded fuels and alkenes [2]. (d) Any balanced alkane $\rightarrow$ shorter alkane + alkene equation [2].

13. 13(a) Yeast, 30-40 °C and absence of oxygen [any 2]. (b) $\text{C}_6\text{H}_{12}\text{O}_6 \rightarrow 2\text{C}_2\text{H}_5\text{OH} + 2\text{CO}_2$ [2]. (c) Fast or continuous and pure product or high yield [2]. (d) Renewable feedstock and low temperature or pressure [2].

14. 14(a) Bromine adds across the double bond and the colour disappears [2]. (b) Addition [1]. (c) The double bond opens to single bonds linking monomers [2]. (d) $[-\text{CH}_2-\text{CH}_2-]_n$ with continuation bonds [3].

15. 15(a) Ethyl ethanoate [1]. (b) Water [1]. (c) Esterification and concentrated sulfuric acid [2]. (d) With metal, base or carbonate, giving correct products [any 2]. (e) Products can react to reform reactants and equilibrium can be established [2].

Self-assessment

Circle the first incorrect step and classify it as knowledge, equation or formula, mole ratio, unit conversion, observation, practical method or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.